VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC301

Digital Signal Processing & Filter Design - Laboratory Manual & Practica

Syllabus Focus: EC301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Discrete Fourier Transform & FFT

" Syllabus Scope: DFT properties, circular convolution vs linear convolution, Decimation-In-Time DIT and

Unit 2: Infinite Impulse Response IIR Filter Design

" Syllabus Scope: Analog Butterworth and Chebyshev filter approximations, Bilinear Transformation, Impuls

EC301: Digital Signal Processing & Filter Design - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Finite Impulse Response FIR Filter Design

" Linear phase conditions, symmetric/antisymmetric impulse responses, windowing methods Rectangular, Hamming,

Unit 4: Digital Filter Realization Structures

" Direct Form I, Direct Form II, Cascade, Parallel, and Lattice ladder realizations for IIR and FIR architectu

Unit 5: Finite Word Length Effects in DSP

" Fixed vs floating-point representations, ADC quantization noise, coefficient round-off error, limit cycle os

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Computation of N-Point DFT and Verification of Circular Convolution in Python

Lab 2: Implementation of Radix-2 DIT-FFT Butterfly Algorithm from Scratch

Lab 3: Design and Magnitude Response Verification of IIR Low-Pass Butterworth Filter

Lab 4: Design and Spectral Analysis of Linear Phase FIR Filter using Windowing Methods

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John G. Proakis, Dimitris G. Manolakis - Digital Signal Processing: Principles, Algorithms, and Applica

* Alan V. Oppenheim, Ronald W. Schafer - Discrete-Time Signal Processing Pearson 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students